CDA 3103

Recitation 2 – Logic Gates

Question 1

- Show that the following statement is true using perfect induction.
- X NAND Y = (X' NOR Y')'

X	Y	X'	Y'	X AND Y	X NAND Y	X' OR Y'	X' NOR Y'	(X' NOR Y')'
0	0	1	1	0	1	1	0	1
0	1	1	0	0	1	1	0	1
1	0	0	1	0	1	1	0	1
1	1	0	0	1	0	0	1	0

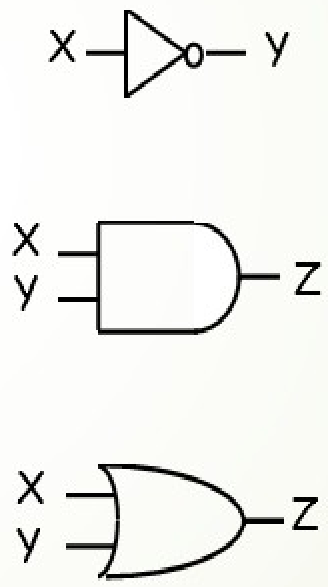
Question 1

- Show that the following statement is true using perfect induction.
- X NOR Y = (X’ NAND Y’)’

X	Y	X’	Y’	X OR Y	X NOR Y	X’ AND Y’	X’ NAND Y’	(X’ NAND Y’)’
0	0	1	1	0	1	1	0	1
0	1	1	0	1	0	0	1	0
1	0	0	1	1	0	0	1	0
1	1	0	0	1	0	0	1	0

Review: Logic Gates

- NOT   X' $\overline{X}$ ~X
- AND   X • Y   X ∧ Y
- OR   X + Y   X ∨ Y



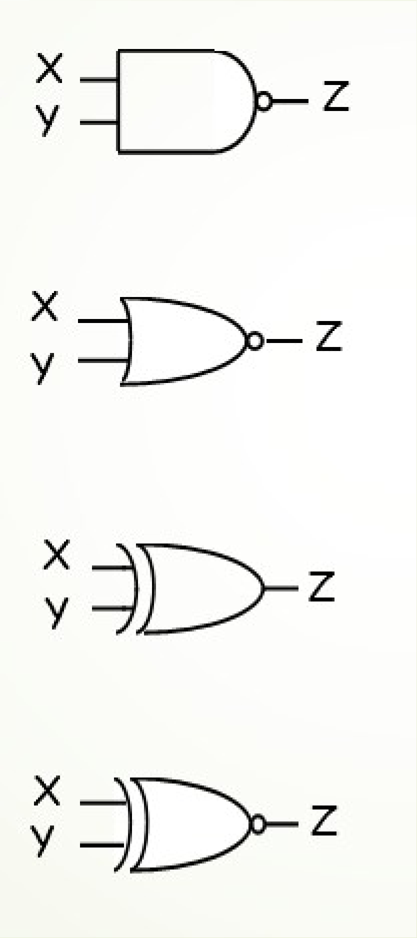
x	y
0	1
1	0

x	y	z
0	0	0
0	1	0
1	0	0
1	1	1

x	y	z
0	0	0
0	1	1
1	0	1
1	1	1

Review: Logic Gates

- NAND
- NOR
- XOR
  X ⊕ Y
- XNOR
  X ⊕ Y



X	Y	Z
0	0	1
0	1	1
1	0	1
1	1	0

X	Y	Z
0	0	1
0	1	0
1	0	0
1	1	0

X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	0

X xor Y = X Y' + X' Y
X or Y but not both
("inequality", "difference")

X	Y	Z
0	0	1
0	1	0
1	0	0
1	1	1

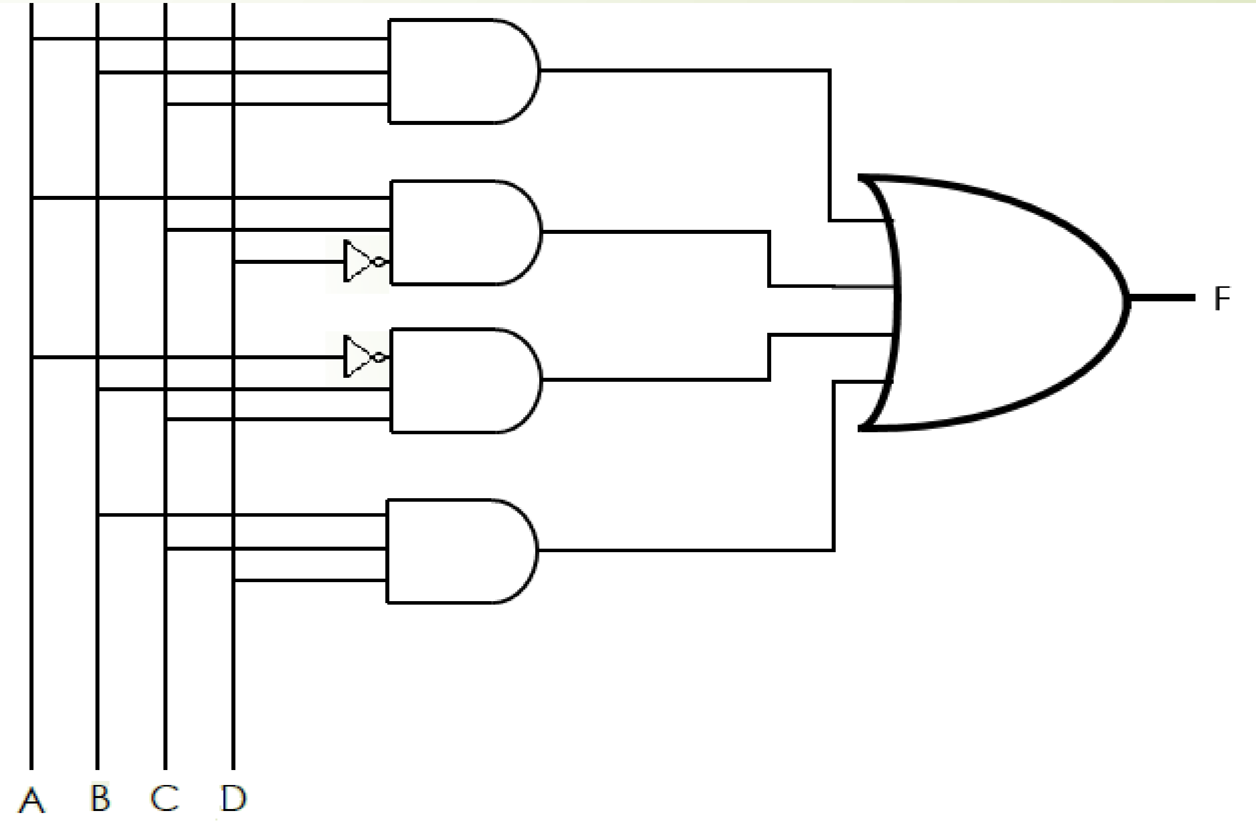
X xnor Y = X Y + X' Y'
X and Y are the same
("equality", "coincidence")

Question 2

- Represent the function F = ABC + ACD' + A'BC + BCD using
  - Any combination of logic gates
  - Only NAND gates
  - Only NOR gates

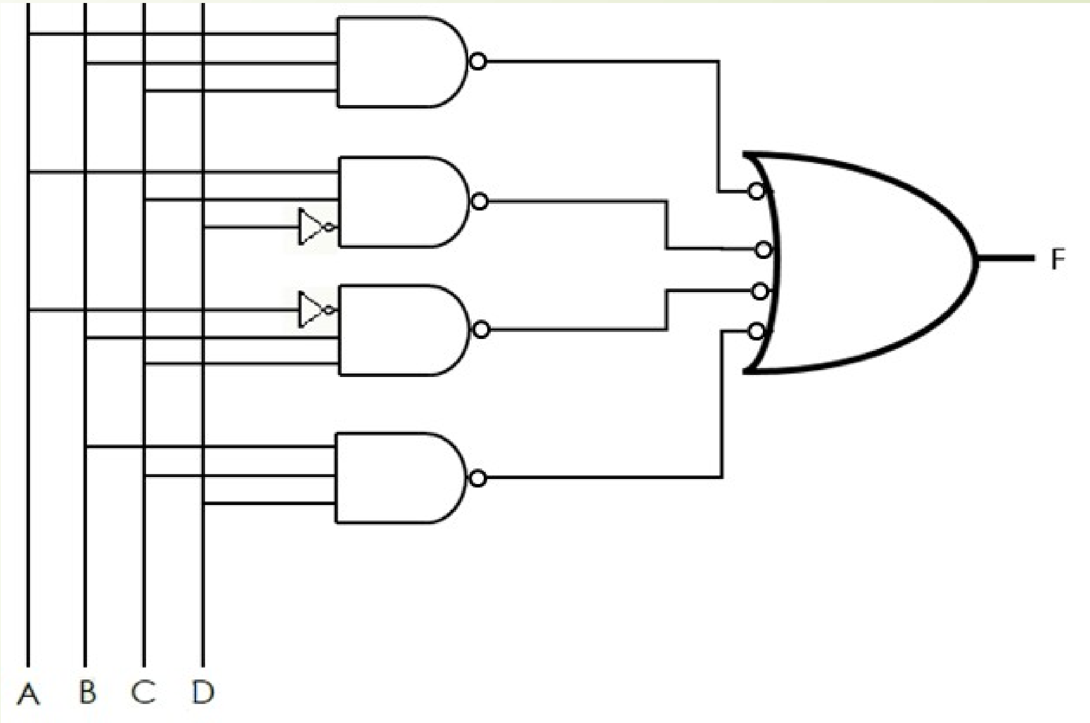
Question 2.1

- Represent the function F = ABC + ACD' + A'BC + BCD using any combination of logic gates



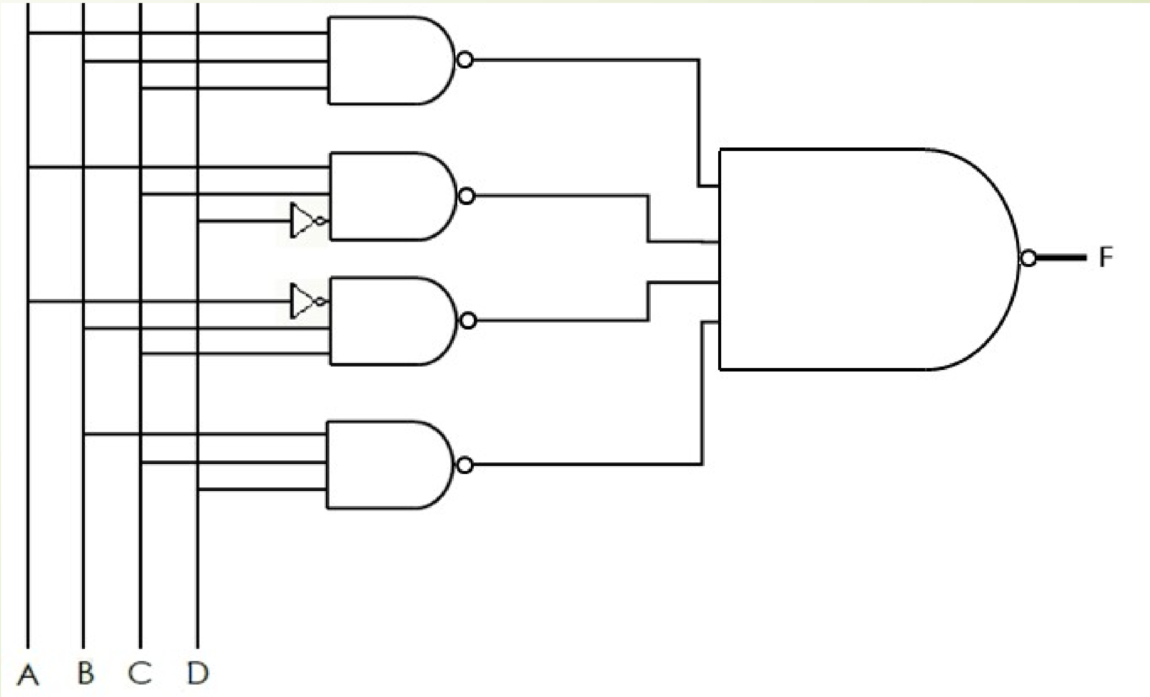
Question 2.2

- Represent the function F = ABC + ACD’ + A’BC + BCD using only NAND gates
  - Step 1: Convert AND gates to NAND gates by inverting the output of the AND gates and the input of the OR gate.



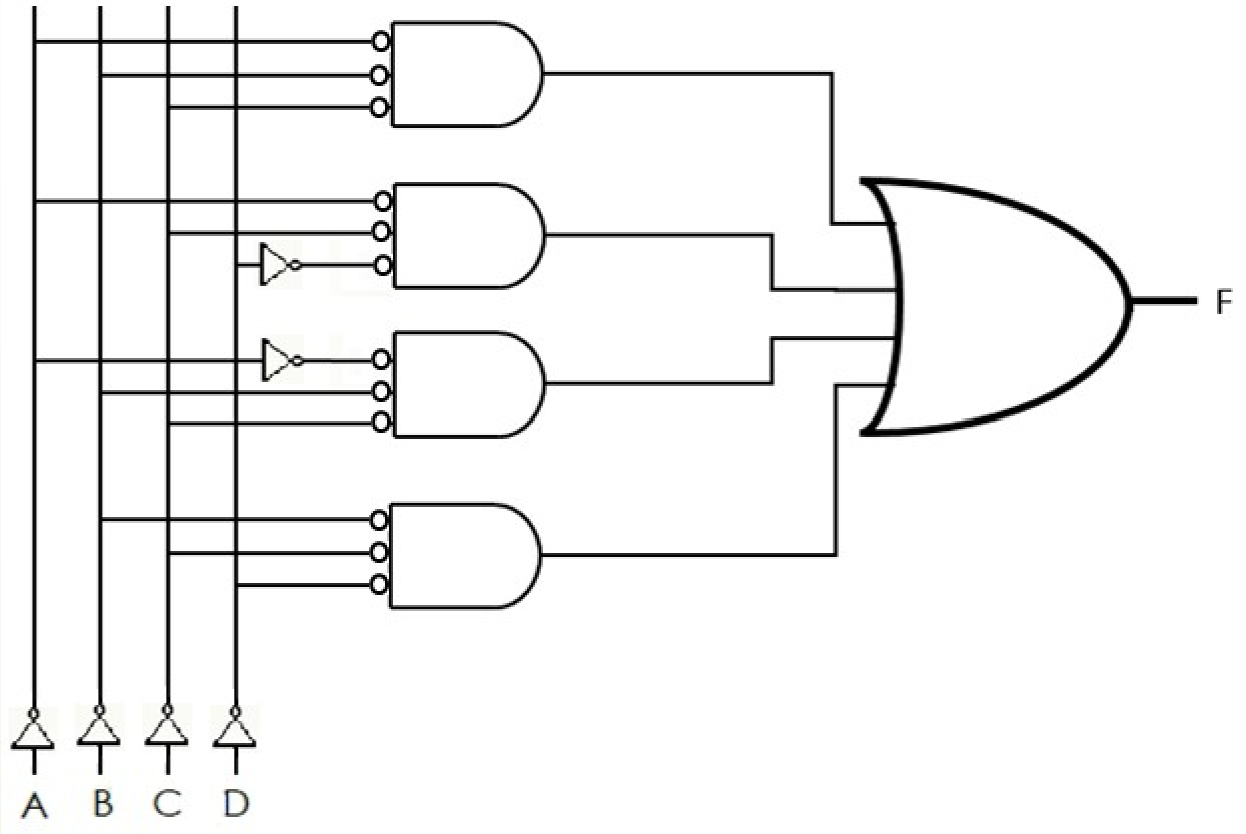
Question 2.2

- Represent the function F = ABC + ACD’ + A’BC + BCD using only NAND gates
  - Step 2: Convert the OR gate with inverted inputs to a NAND gate.



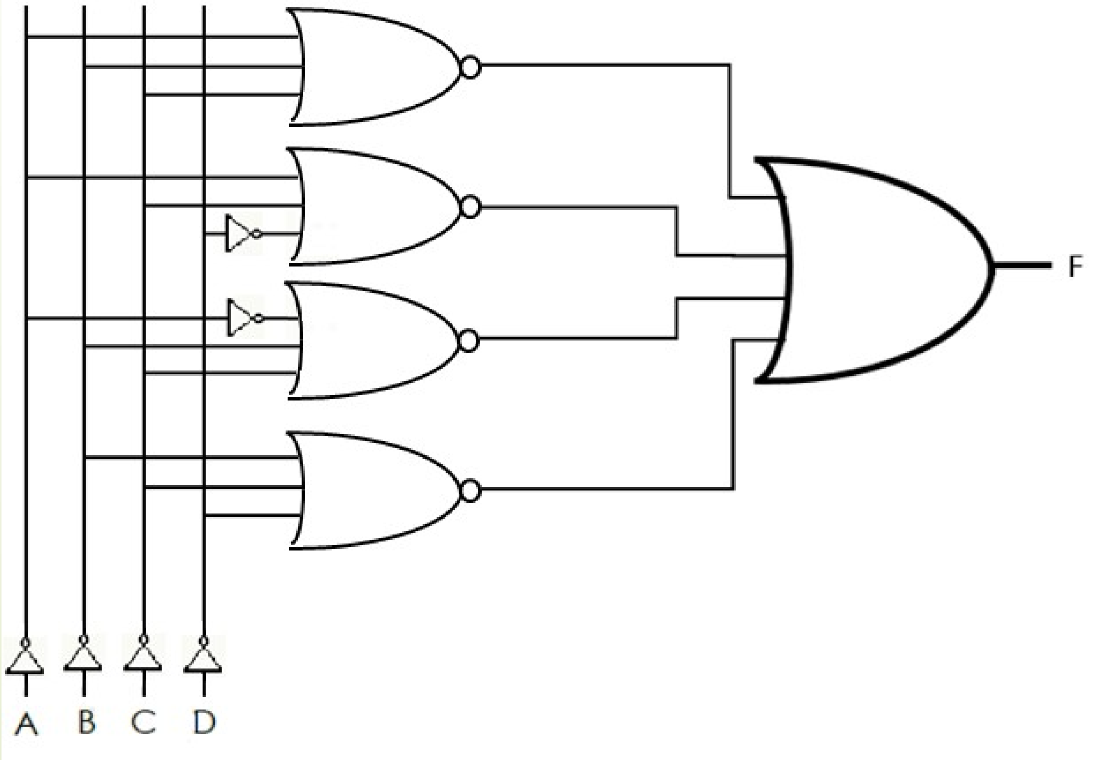
Question 2.3

- Represent the function F = ABC + ACD’ + A’BC + BCD using only NOR gates
  - Step 1: Invert all input signals and the inputs to the AND gates



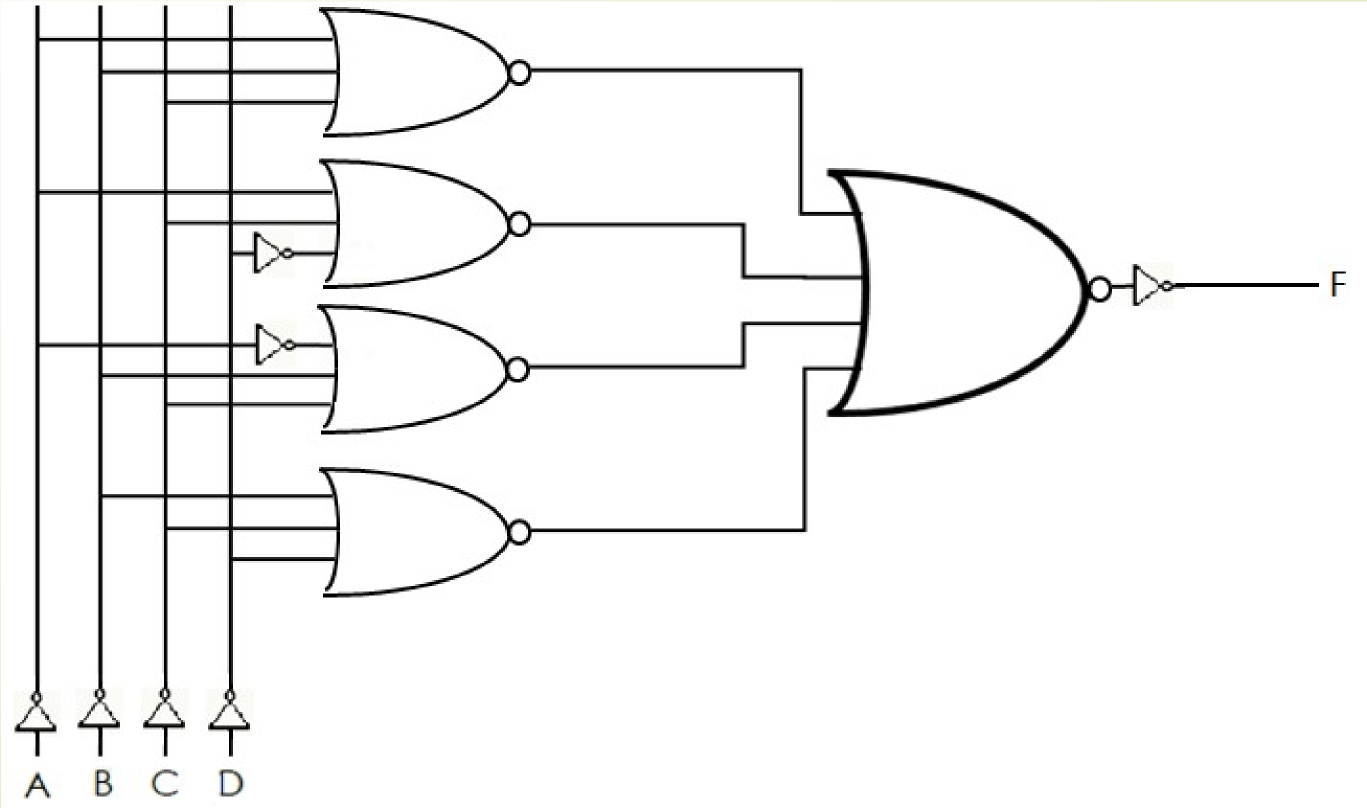
Question 2.3

- Represent the function F = ABC + ACD’ + A’BC + BCD using only NOR gates
  - Step 2: Convert the AND gates to NOR gates



Question 2.3

- Represent the function F = ABC + ACD’ + A’BC + BCD using only NOR gates
  - Step 2: Convert the OR gate to a NOR gate and add an inverter



Question 3

- Consider the following truth table. Identify the function represented by the truth table and reduce it. Show the gate implementation for the original and the reduced function in AND, OR, and NOT gates

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Question 3

- Consider the following truth table. Identify the function:
  - F = A'BC + AB'C' + ABC' + ABC

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

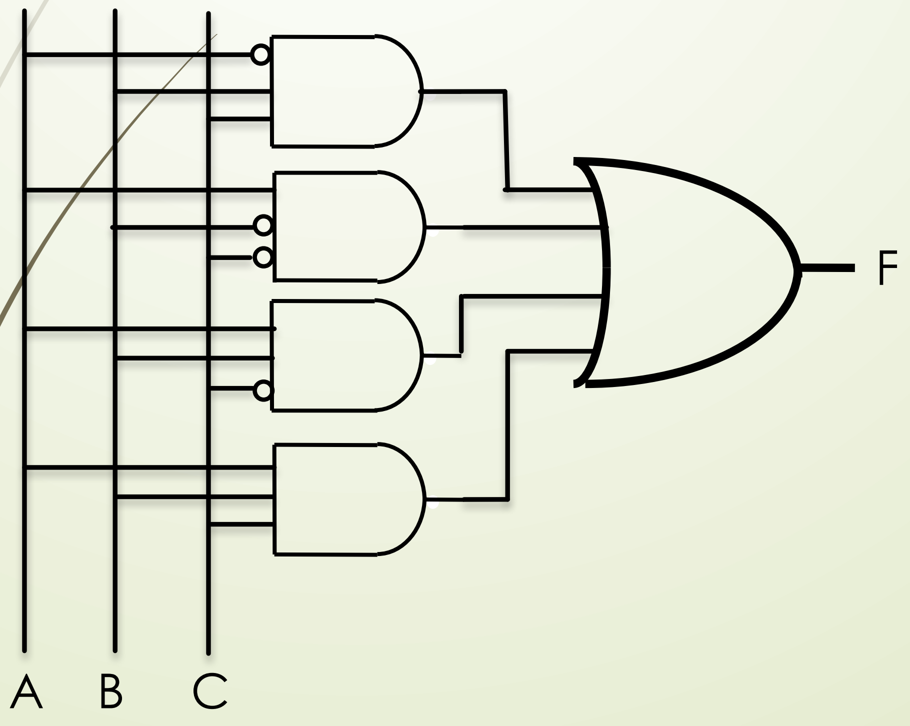
Question 3

- Reduce the function:
  - F = A'BC + AB'C' + ABC' + ABC

= A'BC + ABC + AB'C' + ABC'	Commutative
= (A' + A)BC + AB'C' + ABC'	Distributive
= (1)BC + AB'C' + ABC'	Inverse
= BC + AB'C' + ABC'	Identity
= BC + (B + B')AC'	Distributive
= BC + (1)AC'	Inverse
= BC + AC'	Identity

Question 3

- Show the gate implementation for each function.
  - F = A'BC + AB'C' + ABC' + ABC



  - F = BC + AC'